[Your Project Title Here]

Image Processing — Final Project Report

First Author Name (Student ID), Second Author Name (Student ID), …

I. Problem Description

Clearly describe the image-processing problem your project actually solved. Begin with one or two sentences of real-world motivation, then narrow down to the specific task. Make the input and desired output explicit. For example: "Given a low-light RGB image as input, the system outputs an enhanced image with improved brightness, clearer details, and reduced noise."

Motivation. Explain why this problem matters and where it may be applied, such as medical imaging, traffic monitoring, document analysis, image restoration, industrial inspection, or other real-world scenarios.

Problem Formulation. Clearly define the input, output, and task. For example, the input may be an RGB image, grayscale image, low-resolution image, or noisy image; the output may be a segmentation mask, bounding box, denoised image, enhanced image, reconstructed image, or classification label.

Challenges. Point out what makes the task non-trivial, such as noise, low contrast, illumination changes, blur, occlusion, scale variation, complex backgrounds, or limited training data.

Contributions. Briefly summarize what you implemented or improved, such as a complete image-processing pipeline, a comparison between traditional and deep-learning methods, improved preprocessing or post-processing, a collected dataset, or quantitative and qualitative analysis.

II. Method

Describe the technical approach, algorithms, and experimental procedure used in the project. Walk the reader through your pipeline from input to output: input image → preprocessing → feature extraction / model processing → decision / reconstruction → post-processing → final output. A compact pipeline diagram is encouraged.

Pipeline or System Configuration. List the main stages: pre-processing → feature extraction / model processing → decision / reconstruction → post-processing → final output.

Algorithm Design. Name the algorithms you used and justify the choices, such as Gaussian filtering, Histogram Equalization, Canny edge detection, Otsu thresholding, morphological opening / closing, SIFT, HOG, K-means, CNN, U-Net, or other relevant methods. Explain why your method is suitable compared with obvious alternatives.

Tools & Dataset. List the programming language, libraries, and datasets used, such as Python, OpenCV, NumPy, Matplotlib, scikit-image, PyTorch, MATLAB, etc. Describe the dataset name, source, number of images, image resolution, and train / validation / test split if applicable.

Evaluation. Choose metrics based on the task. Image reconstruction / enhancement may use PSNR, SSIM, MSE, or MAE; segmentation may use IoU, Dice Score, precision, recall, or F1-score; detection or classification may use accuracy, precision, recall, mAP, or a confusion matrix. Specify the baseline used for comparison.

III. Experimental Results

Present the actual results obtained in the project. This section should include both quantitative tables and qualitative visual results, with short explanations under figures and tables.

Quantitative Results. Use tables to compare the baseline and the proposed method. Possible columns include Method, PSNR ↑, SSIM ↑, IoU ↑, F1-score ↑, or other metrics suitable for your task. Make sure the numbers are consistent with your experimental setting.

Qualitative Results. Include representative images, such as Input Image, Baseline Result, Proposed Result, and Ground Truth. Explain whether the method preserves boundaries, reduces noise, improves brightness, or reduces false positives / false negatives.

Result Analysis. Discuss when the method works well, when it fails, and the possible reasons behind these observations.

IV. Discussion

Analyze the experimental results instead of only showing figures or numbers. The discussion should include strengths, limitations, and failure cases.

Strengths. Explain in which cases your method performs well, such as preserving object boundaries, being robust to illumination changes, or improving IoU / PSNR / SSIM compared with the baseline.

Limitations. Honestly describe remaining problems, such as failure under low contrast, severe occlusion, small datasets, long training time, or manually tuned parameters.

Failure Cases. Include at least one or two failure cases and explain the possible causes.

V. Conclusion

Summarize what problem the project solved, what methods were used, what the experimental results demonstrated, whether the proposed method improved over the baseline, and how the work could be improved in the future.

References

If you cite papers, datasets, tools, or websites, list them here in IEEE format and cite them in the text as [1], [2], etc.

[1] J. Canny, “A computational approach to edge detection,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 8, no. 6, pp. 679–698, Nov. 1986.